

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (EC)(CSE)

Dec-Jan. 2013-14

EC 705 Advanced Digital Satellite Communication

Date: 01.01.2014, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Define: Thermal control, EIRP. [02]
(b) What do you mean by Mars Orbiter Mission (MOM) of ISRO? What are its objectives? Explain MOM in detail. [05]

- Q - 2 (a) Describe and Compare the MATV and the CATV systems [07]
(b) Describe the TT&C facilities of a satellite communication system. Are these facilities part of the space segment or ground segment of the system? [07]

OR

- Q - 2 (a) What is the meaning of 'Satellite Attitude'? Briefly describe two forms of attitude control. [07]
(b) Explain the meaning of 'Frequency Reuse'. Describe briefly two methods by which this can be achieved [07]

- Q - 3 (a) A receiving system consists of an antenna having a noise temperature of 60 K, feeding directly into a LNA. The amplifier has a noise temperature of 120 K and a gain of 45 dB. The coaxial feeder between the LNA and the main receiver has a loss of 2 dB, and the main receiver has a noise figure of 9 dB. Calculate the system noise temperature referred to input. [07]
(b) An Antenna having a noise temperature of 35 K is connected through a feeder having 0.5 dB loss to an LNA. The LNA has a noise temperature of 90 K. Calculate the system noise temperature referred to (a) the feeder input and (b) the LNA input. [07]

OR

- Q - 3 (a) Explain what is meant by G/T ratio of a satellite receiving system. A satellite TV signal occupies the full transponder bandwidth of 36 MHz and it must provide a C/N ratio at the destination earth station of 22 dB. Given that the total transmission losses are 200 dB and the destination earth station G/T ratio is 31dB/K, calculate the satellite EIRP required. [07]
(b) What do you mean by noise factor and noise figure? An LNA is connected to a receiver which has a noise figure of 12 dB. The gain of the LNA is 40 dB, and its noise temperature is 120 K. Calculate the overall noise temperature referred to the LNA input. [07]

SECTION – II

Q - 4 (a) What is the need of satellite communication? [01]

(b) Define following terms: Cross polarization discrimination, Polarization isolation, [06]
Ascending node, Inclination, Effective path length, FDMA

Q - 5 (a) Give the difference between terrestrial communication system and satellite [05]
communication

(b) Explain Kepler's law of planetary motion [03]

(c) What do you mean by rain attenuation? Explain rain attenuation for circular [06]
polarization.

OR

Q - 5 (a) Write technical notes on LEO, MEO, HEO and GEO satellites. [07]

(b) Calculate rain attenuation for frequency of 12GHz and for horizontal and vertical [07]
polarization, the rain attenuation which is exceeded for 0.01% of the time and in any
year, for a point rain rate of 10mm/h. The earth station altitude is 600m, and the
antenna elevation angle is 50 degrees. The rain height is 3 km,
($a_h=0.0188$, $a_v=0.168$, $b_h=1.217$, $b_v=1.2$ for $f=12\text{GHz}$)

Q - 6 (a) What do you mean by quantization and modulation? Explain TDMA using a [07]
reference station for burst synchronization

(b) Explain Global Positioning Satellite system [07]

OR

Q - 6 (a) State all the multiple access techniques. Explain any one multiple access technique. [07]

(b) Explain Very Small Aperture Terminal system. [07]
